Part 18: Hypothesis Testing

Summary

Statistical hypothesis testing is a formal framework for “making discoveries.” It is ubiquitous is research, and overused. Nevertheless, it can be useful, and it is definitely important to understand the concept.

In particular, one will see repeated references to p-values when reading research. These are often misinterpreted.

Broadly speaking, hypothesis testing gives us a basis to make a claim that an observed “feature” is “real” and not simply an artifact that occurred due to random variability in our sample.

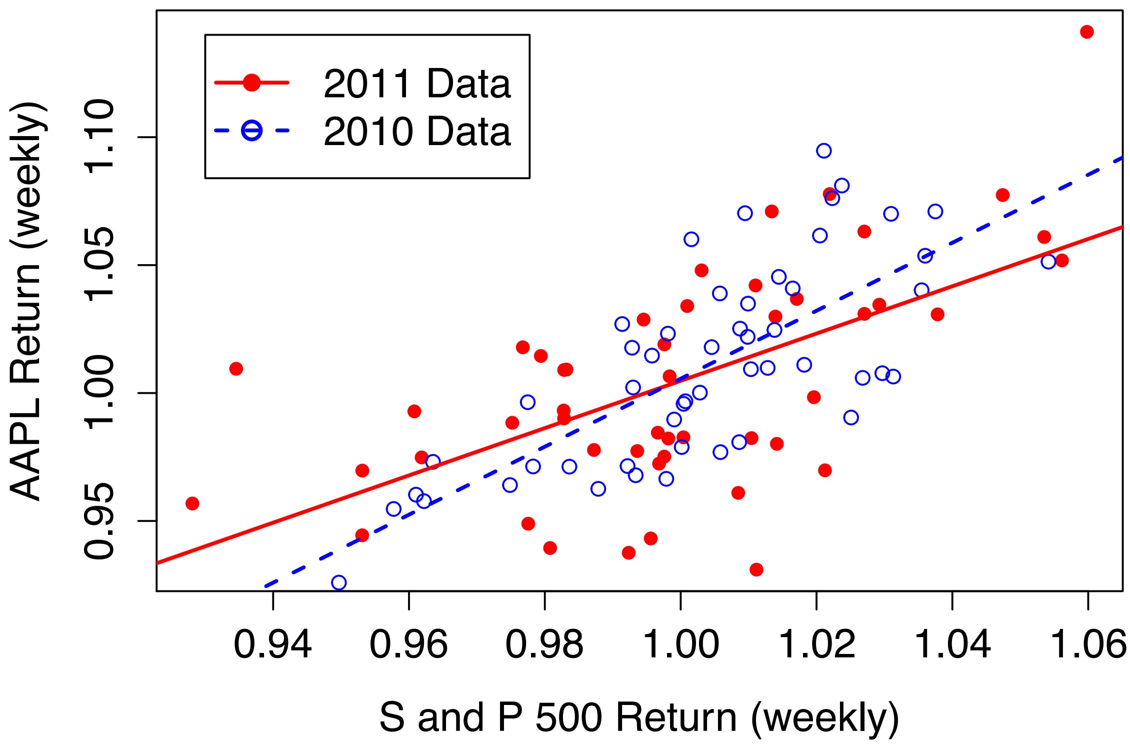
Consider the questions on the following three slides. In each case, the question is not asking for the best estimate of a parameter, but instead asking if the data provide sufficient evidence for some claim.

What do these examples have in common?

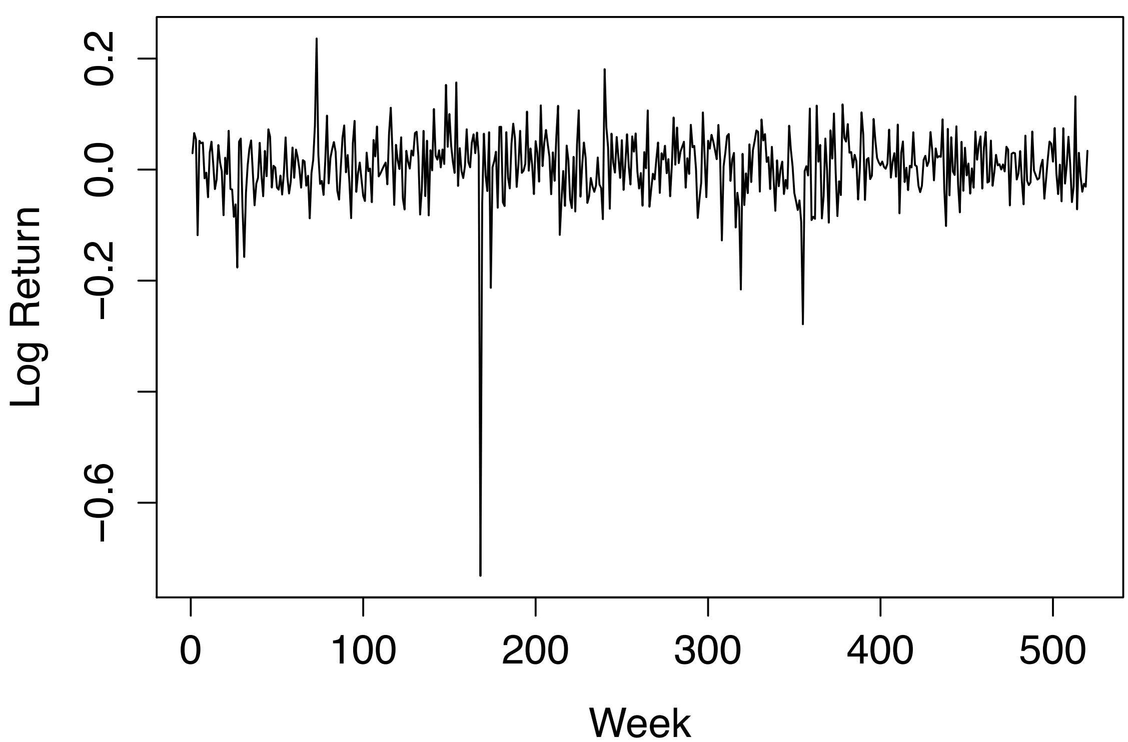
1. They each ask about a feature of the **population** from which the **sample** (observed data) is drawn.

2. The answer to each question is definitely “No.” The issue is whether or not the data provide **sufficient evidence** that the answer is “No.”

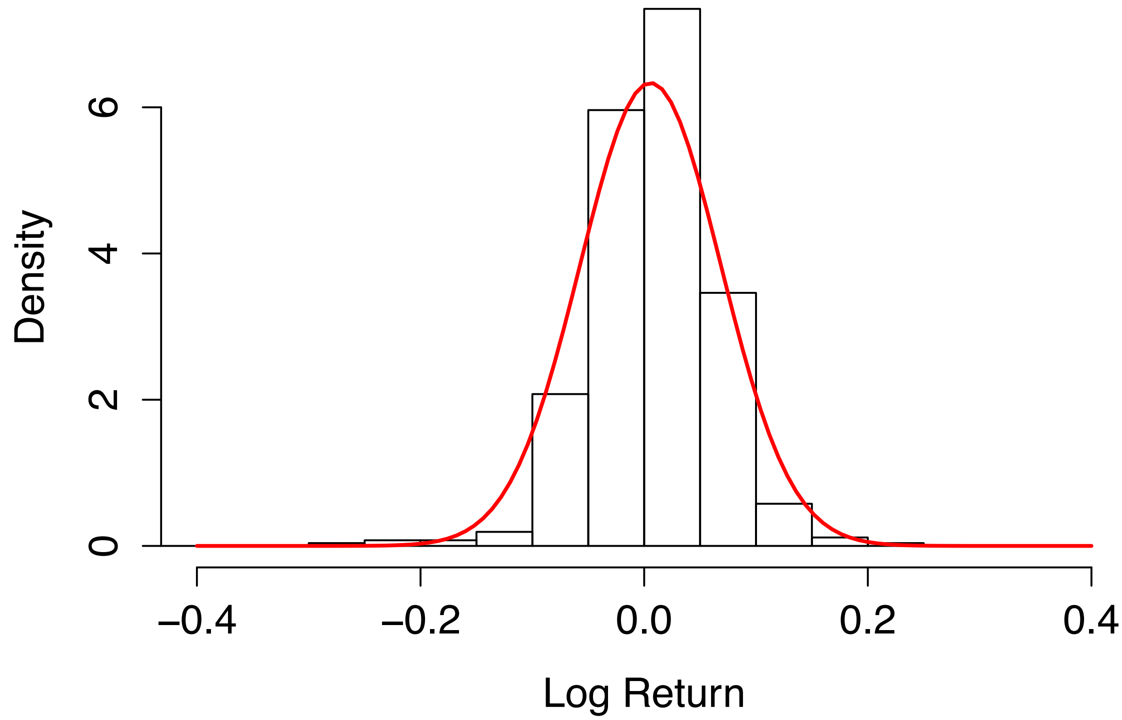
3. Each question puts forth a relatively simple “null hypothesis” and asks “Is there sufficient evidence to reject the null?”

1 

2 Is the slope of the regression line the same in 2010 and 2011?

1 

2 Is the volatility constant over this time period?



Are these data normally distributed?

Such questions as these can be addressed using statistical hypothesis testing.

The key requirement is that the question of interest, "Is there sufficient evidence that. . . ," can be turned into a question regarding the population.

Often the hypothesis will take the form of a question regarding $\boldsymbol{\theta}$, but this is not always the case.

Null and Alternative Hypotheses

The first two key ingredients to a hypothesis test are the null hypothesis (H_0) and the alternative hypothesis (H_a or H_1).

The alternative hypothesis is the claim you are testing to see if there is strong enough evidence to support.

The null hypothesis is the "baseline" or "status quo."

The main thing to keep in mind is the following: **Statistical hypothesis tests are designed to determine if there is strong evidence to support the alternative (and reject the null). They are not designed to test if there is strong evidence to support the null hypothesis.**

Exercise: A pharmaceutical company is testing a new treatment for a disease. The current treatment is 40% effective. Explain why a hypothesis test is appropriate in this case. Formulate a reasonable null and alternative hypothesis for the company.

Let p = effectiveness of new treatment

$H_1: p > 0.4$ vs. $H_0: p = 0.4$

Hyp. testing makes sense because company wants strong evidence before going forward with this new treatment because of the risks

Exercise: Write out null and alternative hypotheses for each of the three original examples. Be clear about what is being tested in each case.

1. $\beta_{1,2010}$ = slope in 2010
   $\beta_{1,2011}$ = " " 2011
   $H_0: \beta_{1,2000} = \beta_{1,2011}$ vs. $H_1: \beta_{1,2000} \neq \beta_{1,2011}$ answers
   "Is there strong evidence that the slopes are different?"

2. $\sigma^2(t)$ = vol. at time t
   $H_0: \sigma^2(t) = \sigma^2 \;\forall t$ vs. $H_1: \sigma^2(t)$ varies with t
   "Is there strong evidence that the vol. is not constant?"

3. H_0: pop. is normal
   H_1: pop is not normal
   "Is there strong evidence that the pop. is not normal?"

The Test Statistic and the Rejection Region

Not surprisingly, data are used to test the hypothesis. As before, random variables $X_1, X_2, \ldots, X_n$ are used to model the observable sample.

The test statistic is the function of the data that we will base our conclusion on. Keep in mind that the test statistic is also a random variable.

Write $T = \underline{T(X_1, X_2, \ldots, X_n)}$.

The rejection region is the set of values for the test statistic such that the conclusion is "reject the null hypothesis in favor of the alternative." In other words, when T falls into the rejection region, we conclude that there is strong evidence in support of the alternative.

How is the rejection region chosen?

Consider the two possible errors we can make with a hypothesis test:

1. A Type I error is made when we reject the null even though the null is true.
2. A Type II error is made when we fail to reject the null when the alternative is true.

The assumption is that we are **primarily concerned with** not **making a Type I error.** We want to limit the probability of making a Type I error.

We form the rejection region so that the probability of a Type I error is less than or equal to some chosen amount α. This is often called the "size" or "level" of the test.

$100(1-\alpha)\%$

Exercise: Suppose that $X_1, X_2, \ldots, X_n$ are i.i.d. Normal(μ, σ^2), with σ^2 known. We want to test $H_0 : \mu = 10$ versus $H_1 : \mu > 10$. A natural test statistic in this case is $\overline{X}$. Formulate the rejection region for a level α test.

$T = \overline{X}$. The rejection should be of the form (c, ∞) for some $c > 10$. How to choose c?

Fix level α. Find c such that

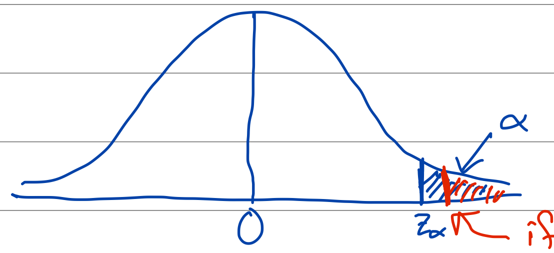
$$P(\text{Type I error}) \leq \alpha$$

i.e., $P(\underbrace{\overline{X} > c \mid H_0 \text{ is true}}) \leq \alpha$

This is when I make a Type I error

$$P\left(\underbrace{\frac{\bar{X}-10}{\sigma/\sqrt{n}}} > \frac{c-10}{\sigma/\sqrt{n}} \;\middle|\; H_0 \text{ is true, i.e. } \underline{\mu=10}\right) \le \alpha$$

$$P\left(Z > \frac{c-10}{\sigma/\sqrt{n}}\right) \le \alpha \qquad Z \sim N(0,1)$$



if c chosen so that $\frac{c-10}{\sigma/\sqrt{n}}$ is here, then $P\left(Z > \frac{c-10}{\sigma/\sqrt{n}}\right) \le \alpha$

need $\boxed{\frac{c-10}{\sigma/\sqrt{n}}} \ge z_\alpha$

i.e. $c \ge 10 + \underbrace{z_\alpha \sigma/\sqrt{n}}_{\text{tolerance}}$

As $n \nearrow$, the tolerance gets smaller

As $\alpha \nearrow$, $z_\alpha \searrow$, so tolerance gets smaller

"Power" $= 1 - P(\text{Type II error})$

$= 1 - P(\bar{X} \le c \mid \mu = ?)$

"If $\mu = 12$, I want $\underline{\text{Power} \ge 0.9}$.

What $\underline{n}$ is needed to achieve this while keeping $P(\text{Type I error}) \le 0.05$"